The Annals of Regional Science (2025) 74:16
https://doi.org/10.1007/s00168-024-01344-4

ORIGINAL PAPER

The shifting relationship between educational attainment and poverty: analysis of seven deep southern states

Amadeo F. Panyi[1] · **Brian E. Whitacre**[1] · **Alicia Young**[1]

Received: 3 October 2023 / Accepted: 1 November 2024 / Published online: 22 January 2025
© The Author(s) 2024

Abstract
Investment in both public and private education is aimed at achieving socioeconomic goals, including the reduction of poverty. While a large body of evidence has established a clear negative relationship between education and poverty, policymakers are often in disagreement about the returns to different levels of educational attainment (for example, bachelor's vs. associate degrees). In this study, the relationship between education and poverty is examined to determine variations over time and space. Census tract-level data were compiled from the 2010–2014 and 2017–2021 5-Year American Community Survey for seven "Deep South" states, widely acknowledged as the poorest region in the USA. Multiscale geographically weighted regression results reveal a shift in the education–poverty relationship from a local scale in the earlier period to a more global relationship in recent years. The results indicate significant shifts in how poverty interacts with educational attainment over time, with associate degrees becoming less influential, while the relationship with bachelor's degrees expands dramatically. These results highlight the need to consider local and temporal context when considering investments in education. The effects of other economic and demographic factors are also reported.

Mathematics Subject Classification I25 · O43 · R11

1 Introduction

Education and human capital have been identified as essential components for economic growth and poverty reduction. Previous studies considered the impact of investments in both education and human capital for social and economic development (Hout 2012; Jepsen et al. 2014). Increased investments in education, both public and private, are geared toward achieving social and/or economic goals. One such socioeconomic goal is the reduction in poverty, a phenomenon that continues

✉ Brian E. Whitacre
brian.whitacre@okstate.edu

[1] Oklahoma State University, 346 Ag Hall, Stillwater, OK 74074, USA

to challenge the development of human society (Padda and Hameed 2018). Past research identified increased educational access and attainment as major factors linked to poverty reduction (Belfield and Bailey, 2001; Jepsen et al. 2014). Increased educational access and attainment reduce poverty due to higher incomes associated with advancement of knowledge and the acquisition and development of skills (Awan et al. 2011). However, this relationship may not hold for all geographical regions. For example, several studies have found that most jobs in areas where agriculture is a major sector require little education, and that having more education does not translate to increased income (Buchmann and Brakewood 2000; Colclough et al. 2000). Nevertheless, the increase and stability in income are generally reported as direct effects of education (Awan et al. 2011; Morgan and David 1963).

In the USA, education has been viewed as a path out of poverty to the extent that elementary and secondary education has been a free public good (Tierney 2015; Strumbos et al. 2018). This is evidenced by federal and state governments' investment in educational institutions and policy initiatives aimed at increasing educational access and attainment (Ladd 2012; Jepsen et al. 2014; Strumbos et al. 2018). Even so, poverty continues to be a burden that destabilizes social and economic progress, making it a continued subject of study. A study by Rank et al. (2021) revealed that about 60% of Americans between the ages of 20 and 75 will experience living below the poverty line for at least a year, while about 75% will encounter either poverty or near poverty. Consequently, individuals from poor households struggle to earn a college degree, while low-income students are less likely to pursue post-secondary education (Haskins et al., 2009; Strumbos et al. 2018).

Although past research has evaluated the impact of education on income distribution (Morgan and David 1963; Ladd 2012; Jepsen et al. 2014), the current literature on education and poverty in the USA lacks empirical evidence on the effects of different educational attainment levels on poverty. Earlier studies have found consistent evidence of positive earnings and lower poverty rates associated with associate and bachelor's degrees (Bailey et al. 2004; Baum et al. 2010; Jepsen et al. 2014; Dadgar &Trimble, 2015; Ma et al. 2016; Strumbos et al. 2018). For the second quarter of 2024, median weekly earnings for full-time workers with a bachelor's degree were $1684 while those with some college or an associate degree earned $1057—both notably higher than the $916 earned by those with high a school diploma (Bureau of Labor Statistics 2024a, b). Researchers have found evidence of both economic and non-economic benefits of post-secondary education. Better health outcomes, reduced welfare receipt, and decreased criminal involvement are associated with additional years of education ranging from community colleges to four-year degrees (Belfield and Baily 2011; Chan 2016).

Of particular importance are the questions of what educational attainment level matters and whether the effects of education on poverty vary over time and space. Some researchers and policymakers call for increased investment in associate degree programs, while others advocate greater investments in bachelor's degrees. For example, Strumbos et al. (2018) propose a national policy based on the Accelerated Study in Associate Programs (ASAP) due to its greater degree completion rate and lower cost for associate degree programs relative to four-year degrees. A recent article by Tough (2023) published by the New York Times

Magazine documents that the percentage of young adults who believe that a college degree is important declined from 74% in the early 2010s to only 41% in more recent years. Further, 45% of young Americans in the current generation believe that having a high school diploma alone is sufficient to guarantee financial stability. Tough (2023) also reports that, unlike parents from a decade ago who prioritized college education, 50% of American parents today would rather their children *not* attend a four-year college. A possible reason for such outcomes may be American's skepticism about the value and cost of a college degree. A recent article by Fry et al. (2024) published by Pew Research Center documents that about 47% of adults believe a four-year college is worth the cost if it can be obtained without taking loans and that 22% believe that having a four-year college education is worth the cost even if it means taking loans; however, 29% believe that college is not worth the cost. Many Americans compare salaries in skilled trades (plumbers, pipefitters, steam workers, etc.) which do not require a four-year college education against the rising costs of college education and the associated student debt. For example, the median annual salary of plumbers, pipefitters, and steamfitters with high school diploma or equivalent as the typical entry-level education was $61,550 in 2023 (Bureau of Labor Statistics, 2024b), while the average student loan debt (including private loan debt) is $40,681. This contrast fuels skepticism about whether the return on investment for a college education justifies the financial burden. Nevertheless, the gap in earnings between adults with and without a college degree continues to widen (Fry et al. 2024). In light of this evidence, the relationship between education and poverty should be examined to identify structural changes and potential spatial variations. Does the evidence suggest that high school or associate degrees have become more important for reducing poverty? Or is the opposite true?

This study examines the relationship between education and poverty in seven states typically considered as comprising the Deep South region of the USA—Alabama, Florida, Georgia, Louisiana, Mississippi, Texas, and South Carolina—taking into cognizance any geographical and structural changes in the relationship over the past decade. This region is widely recognized as having high rates of poverty, particularly for levels of "deep" poverty (Lahr 2020; Baker 2022). Census tracts are used as the geographical unit of analysis, with five-year American Community Survey (ACS) data on poverty, education, employment, demographics, and family structure taken from two distinct periods (2010–2014 and 2017–2021). A multiscale geographically weighted regression (MGWR) and a standardized (global) spatial lag regression are estimated for each period. The econometric specification of the MGWR and the analysis of data from two periods enable the estimation (and visualization) of the variations and structural shifts in the relationship between different educational attainment levels and poverty over time and space. Two hypotheses were proposed for this study. First, we hypothesized that the established inverse relationship between educational attainment and poverty demonstrated in many prior studies continues to hold—that is, higher levels in the percentage of the population with an associate or bachelor's degree were associated with lower poverty rates. Secondly, we hypothesized that the effects of associate or bachelor's degree on poverty have shifted over the past decade, perhaps reflecting the broader change in national

attitude about the importance of a college degree. We refrained from hypothesizing about the expected signs of these changes, leaving them as an open research question. We then explored potential mechanisms underlying our findings, including education level impacts on participation in the labor force.

2 Data and methods

To examine the relationship between education and poverty, census tract-level data were compiled from the 2010–2014 and 2017–2021 5-Year American Community Survey (ACS) for the seven southern states in the study. Census tracts were used as units of observation because they provide a relatively stable set of statistical data based on the subdivisions of a county (Gallardo and Whitacre 2018). A major advantage of the census tract data over county level data is that it considers critical dynamics such as the availability of and proximity to towns and roads (Gallardo and Whitacre 2018) and offers finer levels of geographic granularity valuable for studying variations within counties. Tracts are designed to capture local variations in socioeconomic factors and other characteristics, thus providing a more appropriate level of detail which may lead to statistically significant results, as larger units may dilute the impact of specific factors (Cromartie and Swanson 1996). The 2010–2014 ACS data used Census 2010 geographies and included 15,374 tracts across the seven states, while the 2017–2021 data were updated for Census 2020 geographies and encompassed 19,648 tracts after excluding those with missing observations. The dependent variable chosen was the percentage of the population below the poverty level, which is typical in the regional science literature (Weber et al. 2005; Blank 2005). The primary variables of interest include the percentage of the population aged 25 or older with a high school degree or equivalent, an associate degree, a bachelor's degree, or a graduate or professional degree. Summary statistics for the data used in the estimation are provided in Table 1.

Table 1 shows that 28.9% of residents over the age of 25 in the study area had a high school diploma as their highest degree in the 2010–2014 ACS. However, this percentage dropped to 27.8% in the 2017–2021 period: over a percentage point reduction from roughly a decade earlier. Alternatively, the percentage of population with an associate degree (+1.2 percentage points), bachelor's degree (+2.4), or graduate degree (+1.9) increased during this time. These are similar to the changes seen for the nation. The increasing prevalence of associate degrees may be attributed to factors such as the shorter period to complete / lower student debt associated with pursuing an associate degree relative to a four-year college degree, the practical on-hands training provided in a specific field, or the flexible scheduling offered, among other factors. However, the largest change in educational attainment seen over this period was for bachelor's degrees, perhaps because it is still viewed as the most reliable route to the middle class (Carnevale and Simth, 2022). This increase in population with a bachelor's degree may be driven by both demand and supply forces. On the demand side, there has been a notable shift in the labor market toward industries requiring higher levels of education. The expanding industries in technology, healthcare, energy, and manufacturing are driving demand for more skilled and educated

Table 1 Summary statistics. *Source* U.S. Census Bureau, American Community Survey

Variable	2010–2014 ACS Census tracts				2017–2021 ACS Census tracts			
	Mean	SD	Min	Max	Mean	SD	Min	Max
High school degree	28.922	10.061	0.400	76.700	27.810	11.424	0.100	75.100
Associate degree	7.343	3.383	0.000	35.200	8.498	4.354	0.000	48.100
Bachelor's degree	16.181	10.372	0.000	66.300	18.559	11.299	0.000	83.300
Graduate degree	9.086	8.072	0.000	69.200	10.986	9.462	0.000	72.000
Poverty rate	19.321	12.881	0.000	90.800	15.525	11.840	0.000	87.400
Unemployment rate	10.208	5.947	0.000	73.000	5.787	4.748	0.000	56.300
PTMC employment	12.560	7.347	0.000	56.200	13.769	8.496	0.000	78.400
AFFM employment	2.535	4.498	0.000	68.400	2.106	4.133	0.000	68.000
FIRERL employment	6.241	4.197	0.000	37.900	6.372	5.009	0.000	59.500
AERAFs employment	10.161	6.181	0.000	82.400	9.725	6.499	0.000	70.000
PA employment	5.020	4.423	0.000	75.500	4.577	4.331	0.000	78.600
Percent married	66.692	13.775	0.000	100.000	46.958	17.313	0.000	100.00
Percent single mother	8.374	8.393	0.000	76.000	6.173	5.808	0.000	76.500
Percent disabled	13.849	5.953	0.000	83.700	13.773	6.544	0.000	87.800
Percent Hispanic	20.626	24.936	0.000	100.000	22.724	25.901	0.000	100.00
Percent Asian	2.553	4.732	0.000	72.300	3.175	6.231	0.000	87.200
Percent Black	21.283	24.989	0.000	100.000	20.660	24.164	0.000	100.00
Commute Out ratio	25.822	18.216	3.208	93.028	42.261	33.345	4.706	100.00
Number of observations	15,374				19,648			

workers. This economic transformation has increased job postings requiring bachelor's degree for roles that previously did not (Deshpande 2021; Zhang et al. 2024). On the supply side, increased access to education is a contributing factor for the growth in the population with a college degree in the deep south. Many people in the deep south may have pursued college education due to greater access to colleges, online programs, state-level incentives, scholarships, and affordable education programs. While the average tract-level poverty rate in the Deep South declined from 19.3% in the earlier period to 15.5% in 2017–2021, this latter value was still markedly higher than the 12.6% rate for the USA during that time.

Figures 1, 2 and 3 show the map of poverty rates and percentages of the population with associate and bachelor's degrees in the Deep South across the two periods. The pockets of high poverty (Fig. 1) are less pronounced in the later period, reflecting the approximately 4-percentage-point decline in Table 1.

Given that map inspection is not sufficiently rigorous to assess significant clusters and tends to be biased toward the identification of patterns, the Moran's I measure of spatial dependence is used to test for evidence of spatial dependence for each of the variables in Figs. 1, 2 and 3. Moran's I compares the values of a variable among geographies close to each other with the values of all other locations (Moran, 1950). A Moran's I value of zero shows a random spatial pattern, while a significant positive or negative value indicates positive or negative autocorrelation, respectively. A

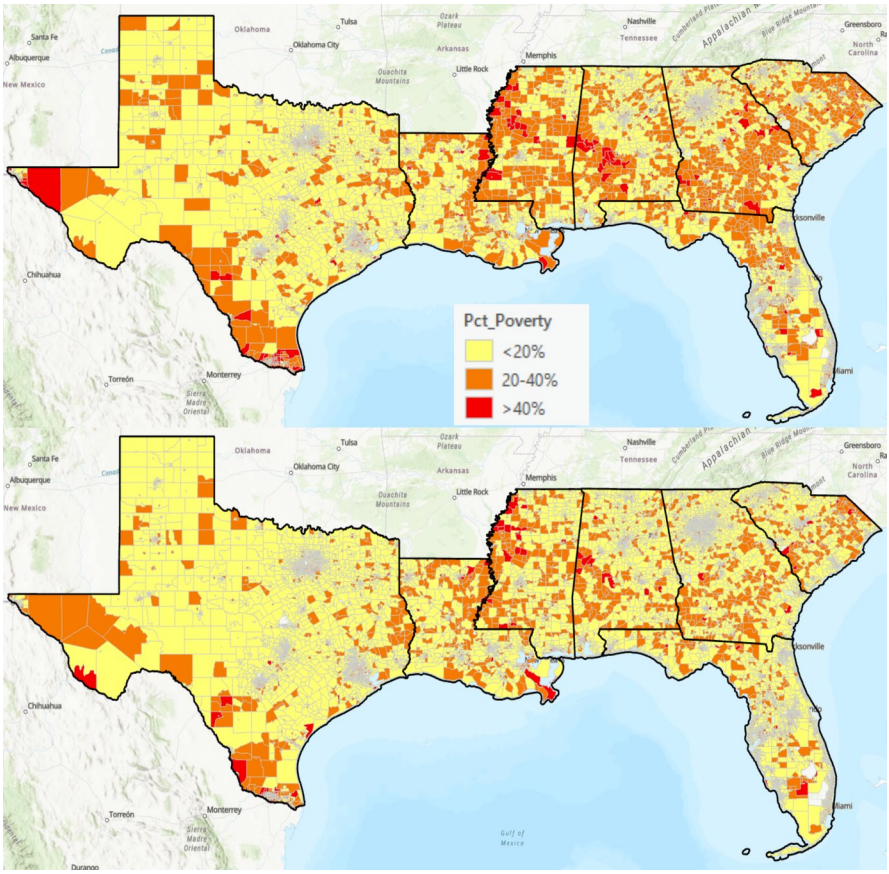

Fig. 1 Poverty rates in the Deep South in 2010–2014 (top) and 2017–2021 (bottom). *Source* American Community Survey, 2010–2014 and 2017–2021 5-year estimates

queen contiguity weight matrix standardized by rows is used. The null hypotheses of random spatial patterns were strongly rejected for all variables indicating significant spatial dependence (Table 2), suggesting that econometric techniques which do not account for the spatial nature of the data may result in biased coefficients (Anselin 2001).

Thirteen control variables were included in the model based on prior research documenting their importance in understanding local levels of poverty (Partridge and Rickman 2007; Partridge and Rickman, 2006; Partridge and Rickman 2005). These include unemployment rates, industrial and family composition, and local sociodemographics such as disability rates and racial and ethnic characteristics. The average unemployment rate for the population 16 years and older was approximately 10.2% with a high of 73% for the 2010–2014 tracts and 5.8% with a high of 56.3% for the 2017–2021 tracts, reflecting a dramatic reduction associated with the easing of the Great Recession. Simple correlation analyses between specific education

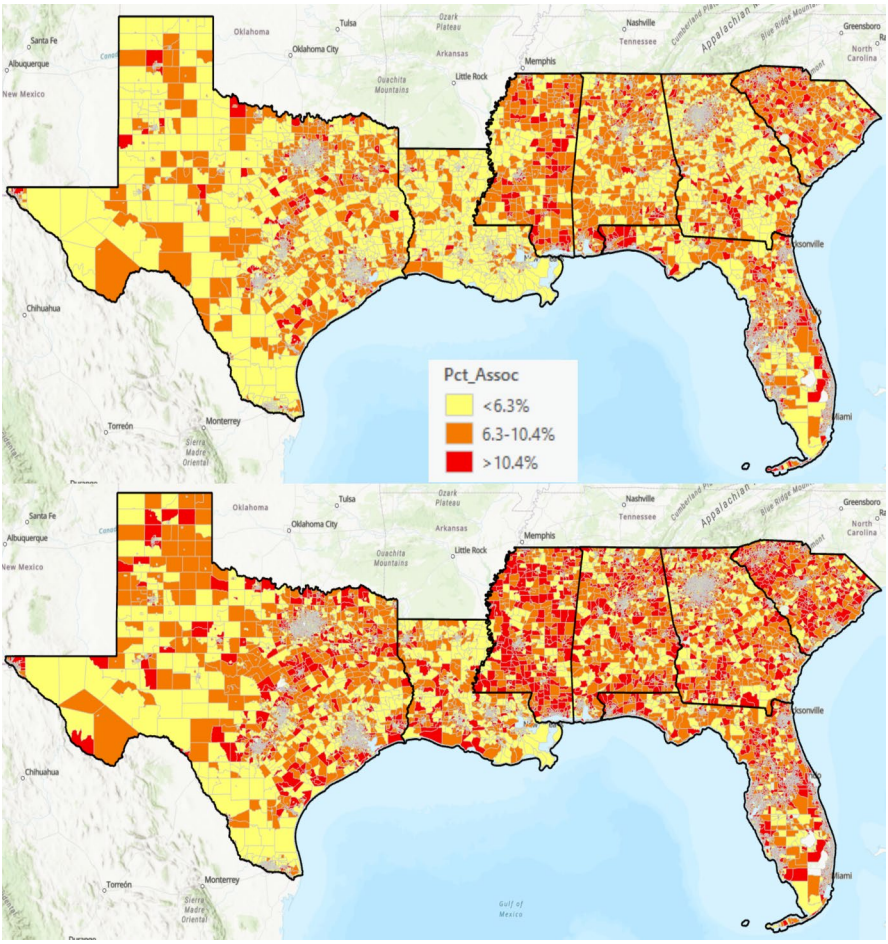

Fig. 2 Percentage of population over 25 with associate degree in the Deep South in 2010–2014 (top) and 2017–2021 (bottom). *Source* American Community Survey, 2010–2014 and 2017–2021 5-year estimates

attainment levels and unemployment reveals relatively strong associations ($r = -0.07$ for associate degree; $r = -0.44$ for bachelor's degrees) for the 2010–2014 tracts and for the 2017–2021 tracts ($r = -0.04$ for associate degree, $r = -0.27$ for bachelor's degrees). Previous studies report similar associations between education and unemployment (Farber 2004; Riddel and Song, 2011; Lavrinovicha et al. 2015). Farber (2004) found that individuals who lose their jobs and possess higher levels of education tend to experience higher rates of post-displacement employment and have an increased likelihood of securing full-time re-employment. Riddel and Song (2011) found evidence that higher post-secondary education reduces the incidence of unemployment. By controlling for unemployment rates, this study more directly captures the relationship between educational attainment and poverty for the percentage of population who are employed.

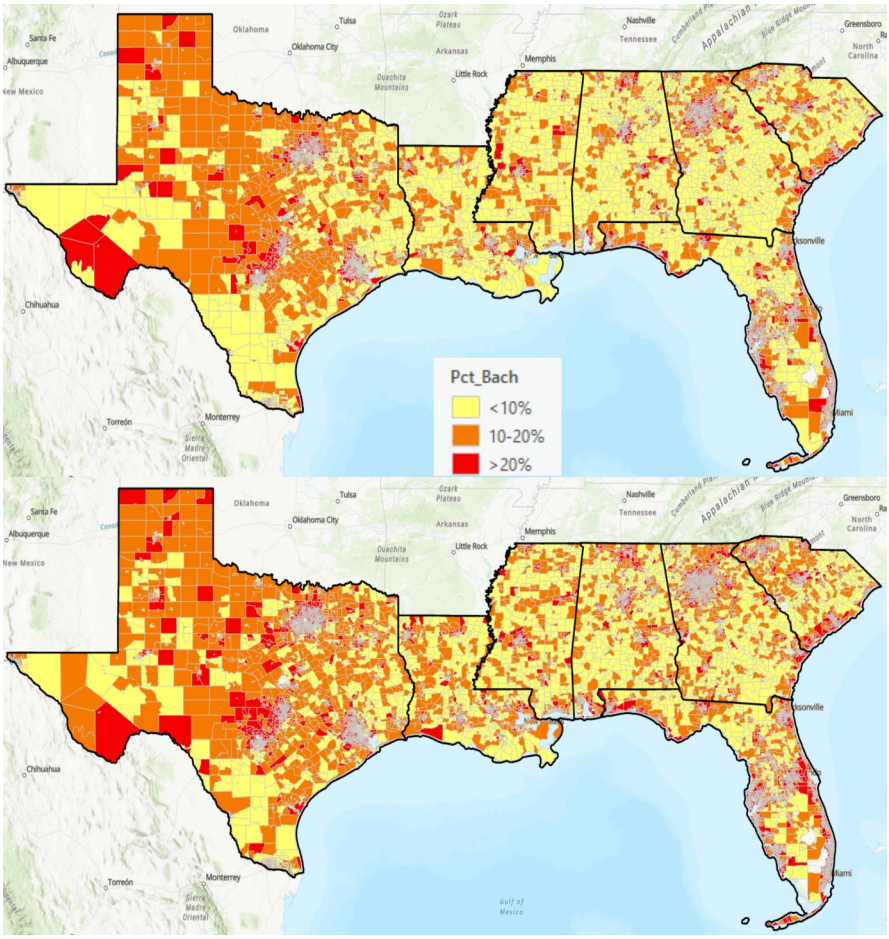

Fig. 3 Percentage of population over 25 with bachelor's degree in the Deep South in 2010–2014 (top) and 2017–2021 (bottom). *Source* American Community Survey, 2010–2014 and 2017–2021 5-year estimates

Table 2 Moran's I test for spatial dependence

Variable	2010–2014 Census tracts		2017–2021 Census tracts	
	Moran's I	*p*-value	Moran's I	*p*-value
Poverty	0.497	0.000	0.430	0.000
Associate degree	0.332	0.000	0.237	0.000
Bachelor's degree	0.649	0.000	0.624	0.000

Past research has considered the effects of the industrial structure on poverty in various types of communities (Albrecht et al., 2000; Levernier et al. 2000; Rupasingha and Goetz 2007). This relationship has been attributed to the adjustment cost facing workers compelled to move between sectors (Levernier et al. 2000). Following this literature, this study estimated the relationship between poverty and employment in five distinct industrial categories: (1) agriculture, forestry, fishing and hunting, and mining (AFFM); (2) production, transportation, manufacturing, and construction (PTMC); (3) finance, insurance, real estate, and rental leasing (FIRERL); (4); arts, entertainment, recreation, and accommodation and food services (AERAFs); and (5) public administration (PA). These industries were chosen based on their importance for the rural and sub-regional economies of the seven-state sample. The average employment rates in PTMC industries were 12.6% and 13.8% for the 2010–2014 and 2017–2021 tracts, respectively, while the average employment rate in AFFM industries was moderately higher (2.5%) in the 2010–2014 tracts relative to 2.1% in the 2017–2021 tracts. Prior research has documented a negative relationship between PTMC employment and poverty (Rupasingha and Goetz 2007) but a positive relationship between AFFM and poverty (Slack et al. 2009).

Family composition is also related to poverty (Atkinson 1992; Cancian and Haskins 2014), as is the local disability rate (Palmer 2011). The average percentage of married-couple households was dramatically lower (−19.7 percentage points) in the later period, while the percentage of single-mother households slightly declined (−2.2). However, the percentage of disabled population was similar between the two periods. Racial and ethnic characteristics are also associated with poverty rates. Greater ethnic diversity often leads to heightened polarization, which can hinder the consensus-building process regarding the allocation of public resources and the development of effective strategies to address persistent poverty and policies aimed at reducing economic growth (Alesina et al. 1999). Along this vein, previous studies have documented significant relationships between ethnic diversity and poverty (Levernier et al. 2000; Rupasingha and Goetz 2007). This study uses the percentages of Hispanic, Asian, and Black population to assess the relationship between race / ethnicity and poverty. Finally, building on the work of Kures and Deller (2023) we include a metric related to tract-level commuting patterns using data from the Census' Longitudinal Origin–Destination Employment Statistics (MacDonald and Pettit 2019). The data show an increasing ratio of workers commuting out of their home tract during this time, reflecting the patterns shown in Kures and Deller (2023).

Previous studies have used different approaches to estimate the returns to different post-secondary education (see Jacobson et al. 2005; Jepsen et al. 2014; Dadgar & Trimble; 2015; Liu et al., 2023). For example, Dadgar & Trimble (2015) used an individual fixed effects model to estimate the returns to wages of distinct community college credentials. Liu et al. (2023) used a spatial autoregressive model (SAR) to identify the effects of education on rural poverty. We begin our analysis with a simple classical ordinary least square (OLS) model specified as:

$$POV_i = \beta EDUC_i + \gamma INDUS_i + \delta DEMOG_i + \varepsilon_i \quad (1)$$

where the dependent variable POV_i is the poverty rate for census tract i; $EDUC_i$, $INDUS_i$, and $DEMOG_i$ are exogenous vectors of educational attainment, industry employment rates, and various demographic characteristics, respectively (all measured at the census tract level); β, γ and δ are vectors of coefficients on the exogenous variables; and $\varepsilon_i \sim N\left(0, \sigma^2\right)$ is an independently and identically distributed random error. This specification may lead to biased coefficients, because OLS is a non-spatial technique and does not account for the spatial autocorrelation documented in Table 2. One approach to overcome this limitation is a spatial lag or autoregressive model that captures the direct effect of poverty from neighboring locations:

$$POV_i = \beta EDUC_i + \gamma INDUS_i + \delta DEMOG_i + \rho WPOV_i + \varepsilon_i \tag{2}$$

where W is a spatial weights matrix (queen contiguity is used in practice) and ρ is the associated spatial coefficient. The spatial lag model accounts for the spatial autocorrelation documented in Table 2 and evidenced in Figs. 1, 2 and 3. This specification allows for the direct effect of poverty from neighboring tracts, reflecting the assumption that poverty rates are not limited by census tract boundaries. The spatial model captures the spillover effects between neighboring tracts, which is critical in our study because poverty rates in a census tracts are likely influenced by poverty rates in neighboring tracts. However, the processes postulated in both (1) and (2) suggest that the response is identical across all census tracts (i.e., β, γ, δ, and ρ do not vary across space), which might not necessarily be the case. Differences in responses among census tracts may render the characterization of the relationship between the variables in the census tracts by a single parameter inaccurate. This is particularly relevant for the topic of this paper, where local relationships with community or four-year colleges likely vary across space.

To better capture the relationship between education and poverty in the Deep South, we used recent advances in geographically weighted regression (GWR) to better address the spatial nature of the relationship. Traditional GWR takes advantage of the spatial dependence in the data to identify the spatial heterogeneity in the relationship between education and poverty—essentially allowing coefficients to vary over space (Brunsdon et al. 2002). However, GWR assumes that all relationships being modeled operate on the same spatial scale and the geographic size of the tracts are homogenous; hence, it fits a single optimal bandwidth as an average of the best bandwidths for each process. This limitation is eliminated by the estimation of the *multiscale* geographically weighted regression (MGWR), which permits covariate-specific bandwidths and is more suitable for areas with diverse geographic sizes. MGWR offers a more appropriate analytical framework due to the diverse geographic sizes of the census tracts used in the study (clearly demonstrated in Figs. 1, 2 and 3, with larger tracts in west Texas, or between rural and urban tracts generally). Specifically, the covariate-specific bandwidths in MGWR yield a more accurate and useful representation of the relationship and are more intuitive (Fotheringham et al. 2017; Li and Fotheringham 2020; Yu et al. 2020). Evidence of the existence of spatial heterogeneity between education and poverty is crucial for policy making—for example, poverty alleviation measures should consider regional educational disparities and documented returns to education in a specific geography.

The multiscale geographically weighted regression used to analyze the spatial heterogeneity in the relation between educational attainment and poverty in the Deep South, estimated using MGWR software (Fotheringham et al. 2017), is specified as:

$$POV_i = \beta_{bi}EDUC_i + \gamma_{bi}INDUS_i + \delta_{bi}DEMOG_i + \varepsilon_i \tag{3}$$

where the main difference with Eq. (1) is the replacement of the global parameter β with β_{bi}, a vector of parameter estimates of bandwidth b for each census tract i (and similarly for γ and δ). The difference between traditional GWR and MGWR is the estimation and use of bandwidth b for each parameter estimate (Fotheringham et al. 2017). This allows the relationships between Y and X to vary at different spatial scales—for example, some might be global in nature while others have smaller bandwidths and only use observations from close neighbors to construct the parameter estimate. Notably, recent improvements in MGWR software allow for the estimation of standard errors and associated test statistics for each geographical unit (Li and Fotheringham 2020). MGWR specifications have started to gain traction in the regional science literature on topics like economic resilience (Watson and Deller 2022), housing values (Sachdeva et al. 2022), and innovation (Karahasan 2023).

3 Results

3.1 Spatial lag results

Table 3 provides the results of the estimated spatial lag regression in Eq. (2) for the 2010–2014 and the 2017–2021 census tracts, in standardized coefficients. Approximately 72% of the variation in poverty is explained by the model for the 2010–2014 census tracts, while 59% of the variation in poverty is explained using the 2017–2021 data. These are higher than the results using traditional OLS (available upon request), which is known to be biased due to the omission of relevant spatial patterns. The results of both time periods reveal varied relationships between the different educational attainment levels and poverty. There is a statistically significant negative relationship between poverty and each educational attainment level in both periods, although the strength of the relationship is reduced in the 2017–2021 specification. For example, in 2010–2014 there is a strong negative relationship between the percentage of individuals with a high school degree and poverty ($\beta = -0.14, p < 0.01$). However, the effect of a high school degree on poverty is moderately reduced in the model estimated with the 2017–2021 tracts ($\beta = -0.07$). The 95% confidence intervals around these point estimates do not overlap, indicating that the difference over time is statistically significant.[1] A similar pattern is observed for associate degree,

[1] The 95% C.I. for High School degree point estimate is [-0.155, -0.122] in 2010–14 and [-0.090, -0.056] in 2017–21.

Table 3 The effect of education on poverty (spatial lag results)

Variable	Standardized (Global) spatial lag coefficient	Significant (*p*-value)	Standardized (Global) spatial lag coefficient	Significant (*p*-value)
	2010–2014		2017–2021	
Constant	0.007*	0.095	0.006	0.227
High school degree	−0.139***	0.000	−0.073***	0.000
Associate degree	−0.117***	0.000	−0.101***	0.000
Bachelor's degree	−0.228***	0.000	−0.191***	0.000
Graduate degree	−0.065***	0.000	−0.058***	0.000
Unemployment rate	0.195***	0.000	0.132***	0.000
PTMC employment	0.009	0.151	0.002	0.747
AFFM employment	0.034***	0.000	0.029***	0.000
FIRERL employment	−0.060***	0.000	−0.040***	0.000
AERAFs employment	0.071***	0.000	0.071***	0.000
PA employment	−0.049***	0.000	−0.033***	0.000
Percent disabled	0.012***	0.000	0.039***	0.000
Percent married	−0.208***	0.000	−0.271***	0.000
Percent single mother	0.249***	0.000	0.151***	0.000
Percent Hispanic	0.069***	0.000	0.029***	0.000
Percent Asian	0.011**	0.014	0.023***	0.000
Percent Black	0.046***	0.000	0.023***	0.000
Commute Out ratio	−0.015***	0.000	−0.019***	0.000
Lagged poverty rate	0.343***	0.001	0.332***	0.000
Adjusted R^2	0.722		0.591	
Number of Observations	15,374		19,648	

*, **, *** significance at the 10%, 5% and 1%, respectively

where a one-standard-deviation increase is associated with a reduction in poverty rates by approximately 0.12 standard deviations in the 2010–2014 model but a 0.10 standard deviation reduction in the effect in the more recent period ($\beta = -0.10, p < 0.01$). However, the 95% confidence intervals around these point estimates do overlap, suggesting that the difference over time is not meaningful.[2] Note that the relative negative effect of an associate degree on poverty was lower than that of a high school degree in the 2010–2014 model but became larger for the 2017–2021 data.

Bachelor's degree attainment had statistically significant and pronounced negative effects on poverty in both models, with each standard deviation increase associated with a reduction in poverty rates by approximately 0.23 standard deviations in 2010–2014. Still statistically significant, this relationship is reduced to −0.19 standard deviations in the model estimated using the 2017–2021 data. These differences over time are statistically significant, suggesting a lowering over time.[3] Previous

[2] The 95% C.I. for Associate degree point estimate is [-0.126, -0.108] in 2010–14 and [-0.111, -0.092] in 2017–21.

[3] The 95% C.I. for Bachelor's degree point estimate is [-0.246, -0.209] in 2010–14 and [-0.208, -0.174] in 2017–21.

studies have reported lower poverty rates and higher earnings associated with bachelor's degree (Baum et al. 2010; Jepsen et al. 2014; Ma et al. 2016; Partridge & Rickman 2007). For example, Partridge and Rickman (2007) reported a negative relationship between the proportion of the population with bachelor's degree and persistently high poverty rates.

Graduate degrees had a statistically significant negative relationship with poverty in either period, suggesting that such degrees offer additional influence relative to a bachelor's degree. However, the impact of graduate degrees was moderate in both models, reducing poverty by approximately 0.06 standard deviation for each additional standard deviation increase. The implication is that while graduate degrees offer additional negative influence on poverty, its relationship has not changed significantly in the last decade. This is confirmed via 95% confidence intervals on the point estimates.[4]

Other control variables behave largely as expected. The spatially lagged poverty rate was positive and highly significant in both periods. The effect of unemployment rates on poverty was strongly positive in both models, with each standard deviation increase associated with an increase in poverty rates by 0.20 standard deviations in the 2010–2014 model and 0.13 standard deviations in the 2017–2021 model. A similar relationship between unemployment and poverty is reported in the literature (Saunders 2002; Tsaliki 2009; Ukpere and Slabert 2009; Šileika and Bekerytė 2013). While no meaningful relationship was observed for the percentage of civilians employed in production, transportation, manufacturing, and construction (PTMC) industries, other industrial employment categories were shown to be influential. The percentage of civilians employed in agriculture, forestry, fishing, and mining (AFFM employment) had a positive effect (i.e., poverty-increasing) on poverty rates, with similar coefficients over time. Research has shown a similar positive relationship between total employment in agriculture and poverty (Albrecht et al., 2000; Levernier et al. 2000; Slack et al. 2009). This effect has been attributed to the predominance of agriculture in rural areas with less diversified economies (resulting in fewer alternative job opportunities) and the elevated levels of underemployment in the agricultural sector relative to other sectors (Slack and Jensen 2004). Also, there was a positive relationship between poverty rates and the percentage of civilians employed in arts, entertainment, recreation, accommodation, and food services (AERAFs) industry—a result that is consistent over time. In contrast to employment in the AFFM and AERAFs industries, employment in the finance, insurance real estate, and rental leasing (FIRERL) industry and employment in the public administration (PA) industry had a negative influence on poverty in both periods. This is consistent with previous studies that have reported a negative relationship between employment in FIRERL industry and poverty rates (Levernier et al. 2000; Slack et al. 2009). A similar pattern is observed for employment in the PA industry, where a standard deviation increase is associated with reductions of approximately 0.03–0.05

[4] The 95% C.I. for graduate degree point estimate is [-0.081, -0.049] in 2010–14 and [-0.073, -0.043] in 2017–21.

standard deviation in poverty rates across both periods. The percentage of disabled population was positively related to poverty rates in both models, but the effect was stronger and much larger during the later period. Previous studies have reported higher poverty rates for people with disabilities (Brucker et al., 2015), and the ACS reports the percentage of disabled population ranging from 12.5% for Texas to 18.5% for Mississippi (ACS, 2022). The findings here reinforce that disability levels are positively associated with poverty rates and suggest that the relationship has strengthened over the past decade.

The percentage of married-couple households had large and negative relationships with poverty in both time periods. A one-standard-deviation increase in the percent of married households reduced poverty by between 0.21 and 0.27 standard deviations in both models; a substantial effect. This is likely attributed to the increased income of the household due to the potential employment of both adults. Thus, the possibility of having two income earners allows married families to enhance their diversification, safeguarding them against volatile labor market conditions that could lead to significant income reductions (Partridge and Rickman 2006). Notably, the higher correlation in the 2017–2021 period comes as the percentage of married households declined by nearly 20 percentage points (Table 1). This shift in family composition—and associated stronger negative impact of marriage on poverty—is an intriguing finding. Alternatively, as the percentage of single-mother households increases, poverty rates increase. This result was consistent across both models and the magnitude of the effects appears to be slightly declining over time. Similar relationships have been reported between poverty and female-headed households (Hanratty and Blank 1993; Levernier et al. 2000; Snyder and McLaughlin, 2004; Snyder et al., 2006).

The relationships between specific racial and ethnic categories and poverty are relatively consistent. The percentage of Asian, Hispanic, and Black population were each associated with increases in poverty in both time periods. Previous studies have found similar significant relationships between race and poverty (Levernier et al. 2000). Interestingly, the positive associations for Hispanic and Black populations decreased over time. Conversely, Partridge and Rickman (2007) found a negative relationship between African-American population and high persistent poverty. This difference in results can be attributed to variations in the time period and units of analysis. There is also a negative relationship between poverty and the percentage of the population that commute to work, with similar impacts in both periods. A one-standard-deviation increase in the percentage of population that commutes out of tract for work leads to an approximately 0.02 standard deviation reduction in both periods. Individuals who commute to other areas for work often do so because those areas offer better job opportunities and higher wages than are available locally. This access to better-paying jobs increases household income, thereby reducing poverty rates.

These spatial lag results provide some level of reference regarding the relationship between poverty and educational attainment across the Deep South. As noted above, however, this model assumes global relationships between each variable and poverty—which likely does not hold given the spatial variation in

Table 4 The effect of education on poverty, 2010–2014 (MGWR)

Variable	Spatial lag coefficient (Table 3)	Median GWR coefficient	MGWR Bandwidths	% positive ($p<0.05$)	% negative ($p<0.05$)
Constant	0.007	−0.026	122.000	4.189	4.709
High school degree	−0.139	−0.103	358.000	0.000	17.548
Associate degree	−0.117	−0.101	710.000	0.000	54.420
Bachelor's degree	−0.228	−0.234	302.000	0.000	49.834
Graduate degree	−0.065	−0.068	342.000	0.000	12.000
Unemployment rate	0.195	0.183	411.000	80.533	0.000
PTMC employment	0.009	0.000	724.000	0.202	1.060
AFFM employment	0.034	0.056	664.000	4.358	0.000
FIRERL employment	−0.060	−0.060	490.000	2.829	9.821
AERAFs employment	0.071	0.067	295.000	9.489	0.000
PA employment	−0.049	−0.065	522.000	0.013	16.780
Percent disabled	0.012	0.029	341.000	3.662	1.392
Percent married	−0.208	−0.218	53.000	0.163	21.847
Percent single mother	0.249	0.250	210.000	75.792	0.000
Percent Hispanic	0.069	0.206	193.000	19.141	0.689
Percent Asian	0.011	−0.002	493.000	3.857	0.000
Percent Black	0.046	0.171	168.000	22.738	0.000
Commute Out ratio	−0.015	−0.014	4797.000	0.000	25.210
Lagged Poverty rate (ρ)	0.343				
Adjusted R^2	0.722	0.805			
Number of Observations	15,374	15,374			

Note: *, **, *** significance at the 10%, 5% and 1%, respectively

educational attainment and poverty rates (Figs. 1, 2 and 3; Table 2). As such, our approach shifts to MGWR to allow these relationships to vary across space.

3.2 Multiscale geographically weighted regression (MGWR) results

Tables 4 and 5 present the results from MGWR models analyzing the relationship between poverty and education (along with relevant controls) for the 2010–2014 and 2017–2021 census tracts, respectively. Both MGWR calibrations utilized an adaptive bandwidth using a bisquare weighting function that allows the degree of locality to change across variables (Fotheringham et al. 2017). The critical t-values for hypothesis testing are adjusted within the MGWR software based on these bandwidths. Each table includes the comparable spatial lag (global) coefficient from Table 3, the median MGWR coefficient and MGWR bandwidth for each variable, and the percentage of census tracts that have statistically positive and negative relationships with poverty at a p-value of 0.05. The covariate-specific bandwidths in the MGWR models indicate how different relationships operate over different spatial scales (Fotheringham et al. 2017).

Table 5 The effect of education on poverty, 2017–2021 (MGWR)

Variable	Spatial lag coefficient (Table 3)	Median GWR coefficient	MGWR Bandwidths	% positive ($p<0.05$)	% negative ($p<0.05$)
Constant	0.006	−0.062	178.000	5.217	11.787
High school degree	−0.073	−0.041	771.000	0.000	2.637
Associate degree	−0.101	−0.078	597.000	0.000	18.775
Bachelor's degree	−0.191	−0.179	875.000	0.000	63.284
Graduate degree	−0.058	−0.073	703.000	0.000	15.304
Unemployment rate	0.132	0.111	403.000	31.438	0.000
PTMC employment	0.002	0.008	1196.000	1.043	0.000
AFFM employment	0.029	0.020	1135.000	1.832	0.000
FIRERL employment	−0.040	−0.018	1509.000	0.000	4.799
AERAFs employment	0.071	0.069	579.000	11.319	0.000
PA employment	−0.033	−0.031	3811.000	0.000	32.176
Percent disabled	0.039	0.035	554.000	6.102	0.000
Percent married	−0.271	−0.284	72.000	0.000	26.303
Percent single mother	0.151	0.128	242.000	25.244	0.000
Percent Hispanic	0.029	0.157	396.000	14.821	2.448
Percent Asian	0.023	0.024	1045.000	0.987	0.092
Percent Black	0.023	0.085	459.000	17.201	0.234
Commute Out ratio	−0.019	−0.009	19,647.000	0.000	20.043
Lagged poverty rate (ρ)	0.331				
Adjusted R^2	0.591	0.670			
Number of Observations	19,648	19,648			

*, **, *** significance at the 10%, 5% and 1%, respectively

The MGWR estimates of the adjusted R^2 demonstrate significant improvement over the spatial lag results. More importantly, the MGWR results reveal that the global parameter estimates assumed by OLS and the spatial lag specification do not hold universally. Moreover, there have been significant variations over time in the geographical extent of where relationships hold. For example, in the 2010–2014 data, associate degree attainment was significantly negatively associated with poverty for 54% of census tracts in the MGWR results. However, in the 2017–2021 data, only 19% of the tracts demonstrate a negative relationship. Conversely, the relationship between obtaining a bachelor's degree and poverty was negative for approximately 50% of tracts in this early period but increases to 63% in the later period. Another interesting (but not related to education) result is the dramatic weakening of the positive relationship between poverty and single-mother status. While this relationship held for 75% of tracts in the earlier period, it was only significant for 25% in the later—possibly due to improved resources for these households during this time (Cruse et al. 2022).

To visualize the variation in how the relationship between educational attainment and poverty has changed over time, Figs. 4 and 5 present the significant (at

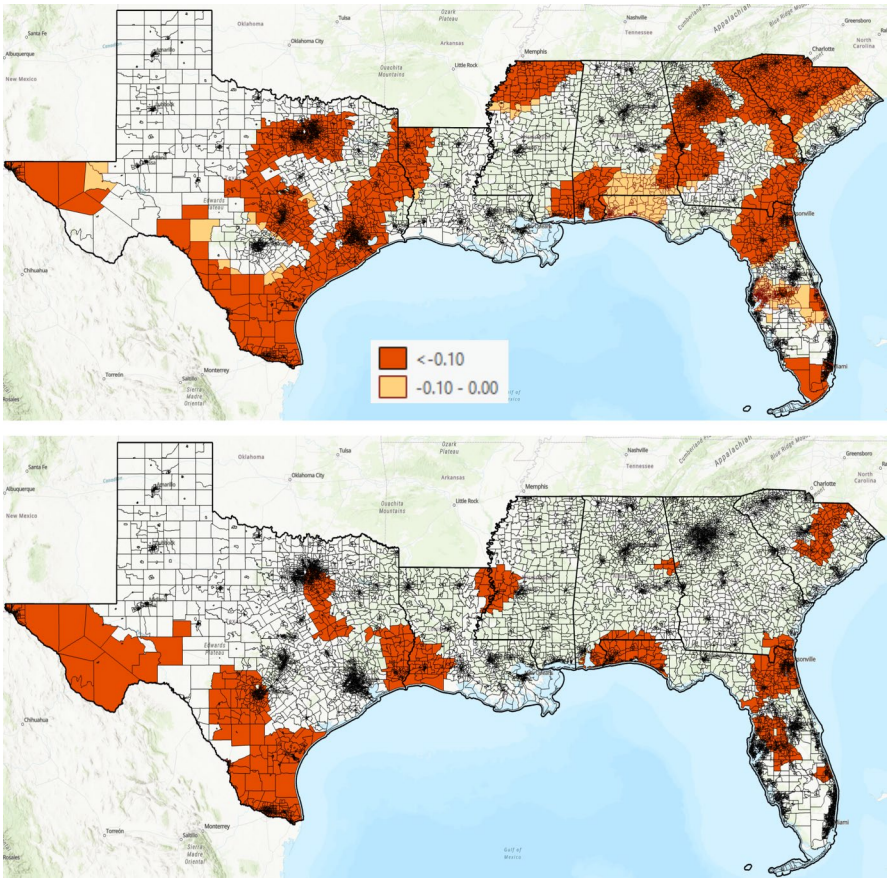

Fig. 4 Multiscale geographically weighted regression coefficients for associate degree: 2010–2014 (top) and 2017–2021 (bottom). Only statistically significant coefficients ($p < 0.05$) are shown

the $p < 0.05$ level) local parameters on associate degree and bachelor's degrees for the 2010–2014 (top) and 2017–2021 (bottom) tracts, respectively. The shifts in these relationships over time are striking, with the negative association between associate degree and poverty rate dramatically *weakening* (Fig. 4) while the association with bachelor's degree became notably stronger—and more universal (Fig. 5).

A closer look at the maps for associate degree coefficients in Fig. 4 shows that higher negative effects are seen in the northern part of Mississippi, South Carolina, and Florida and the north central part of Georgia during the earlier 2010–2014 period. Also, higher negative effects are seen in the southern and southwestern part of Texas, southeastern part of Alabama, and tracts in southern Florida and Georgia. Milder (but still significant) negative relationships are also observed for the southwestern part of Alabama, central Florida, and some tracts in central parts of South Carolina and southern Texas. However, many of these statistically significant

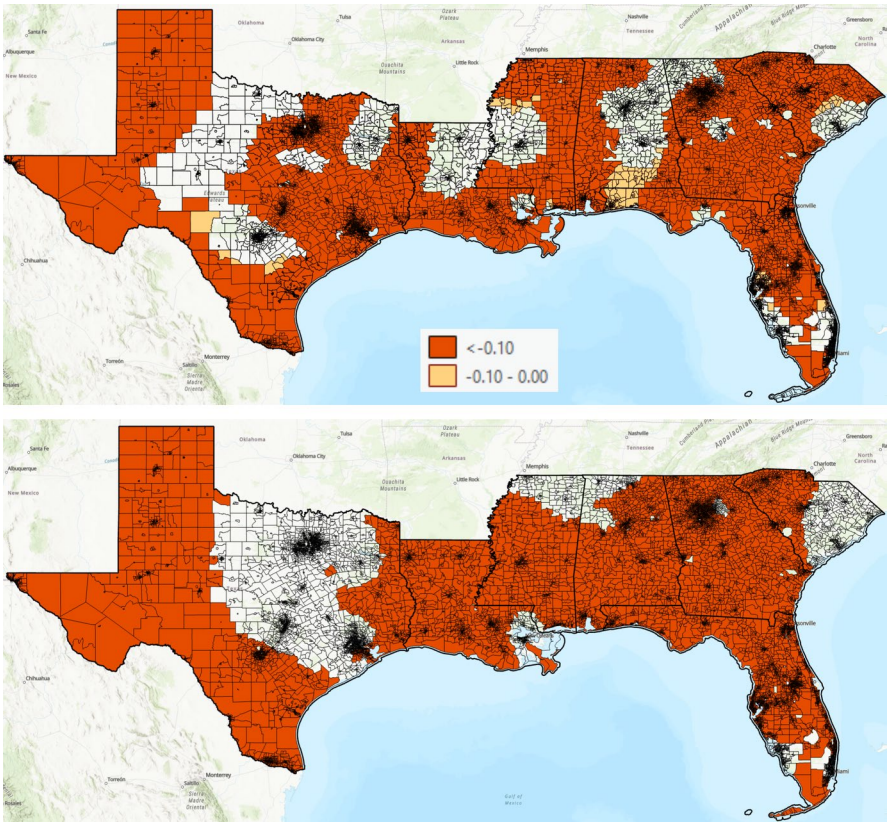

Fig. 5 Multiscale geographically weighted regression coefficients for bachelor's degree: 2010–2014 (top) and 2017–2021 (bottom). Only statistically significant coefficients ($p < 0.05$) are shown

relationships vanish for these tracts in the later period. Notably, the geography of where these results are significant has also changed, with many northern tracts no longer showing any meaningful relationships. These results suggest that even as associate degrees became more common (Fig. 2) they did not necessarily have meaningful impacts on poverty reduction.

These results may also be influenced by the demographic, economic, and educational characteristics. For most places where the influence of associate degrees waned, non-Hispanic whites and Black or African-Americans are the predominant races. Common jobs in many of these counties are sales workers, production and semiconductor processors, labor, freight, and material movers, furniture and related products manufacturing, post-secondary teachers, and cashiers. In addition, some of these tracts are close to major educational institutions offering and awarding four-year and graduate degrees. For example, tracts in the "golden-triangle region" of Mississippi that includes the communities of Starkville, Columbus, and West Point are in close proximity to Mississippi State University, which typically awards over

5000 bachelor's degrees each year (Fallows, 2014). Many of these locations have education as the largest employing industry, and possessing only an associate degree in these locations may not be enough to get a high paying job. In those and similar situations, there might be misalignment between the skills of associate degree holders and the local job market which may lead to unemployment and consequently poverty.

The significant coefficient graphs for bachelor's degree (Fig. 5) tell a markedly different story. In the 2010–2014 map, highly negative relationships are seen for census tracts across most states. Nearly all tracts in Georgia, Florida, and South Carolina exhibit strong negative relationships between poverty and bachelor's degree. Except for a few pockets across the remaining states (Texas, Louisiana, Mississippi, Alabama), most remaining tracts demonstrate statistical significance. In the later period, there is a dramatic shift in the tracts exhibiting strong negative relationship between poverty and bachelor's degree. Most of the tracts that exhibited strong negative relationships in western Texas are not statistically significant in the later period, while some tracts in central Texas did the exact opposite. In the case of Louisiana, nearly all tracts now exhibit a strong negative relationship between poverty and bachelor's degree. The same can be said for Georgia, Florida, and most of Mississippi and Alabama. The main takeaway from Fig. 5 is the dramatic shift over time: *a much broader swath* of census tracts displays strong negative relationships between bachelor's degree attainment and poverty rates in the later period.

A comparison of the bandwidths associated with the educational attainment variables in the MGWR runs (Tables 4 and 5) provides additional context. Most notably, the bandwidth for the associate degree was slightly larger in the earlier period (2010–2014) compared to later period (2017–2021) while the bandwidth for the bachelor's degrees parameter estimates more than doubled over time. The smaller bandwidth for associate degree in the later (2017–2021) period compared to the earlier (2010–2014) period indicates that the relationship has become more local in nature. This is evident in the graph of significant coefficients (Fig. 4) where fewer tracts showed significance in the later period, likely because of these more localized relationships where the potential impact of having an associate degree has a smaller geographic radius. In contrast, the larger bandwidth for bachelor's degree in the later period (2017–2021) relative to the earlier period indicates that the relationship between poverty and bachelor's degree has become global over time. This suggests a more universal (and poverty-reducing) response across the region, perhaps because of broader acceptance of 4-year degrees.

3.3 Potential mechanisms

There are at least three possible mechanisms through which educational attainment might impact local poverty rates. The first is the prevailing wage associated with different educational levels. While we do not have tract-level wage data, broader trends point to stronger gains among bachelor's degrees compared to associate degree holders during this period. The National Center for Educational Studies showed that while annual median earnings showed a statistically significant increase of over

Table 6 Spatial lag regression results of possible mechanisms

Variable	Standardized (Global) spatial lag coefficient	Significant (*p*-value)	Standardized (Global) spatial lag coefficient	Significant (*p*-value)
Labor Force Participation	2010–2014		2017–2021	
Associate degree	0.101***	0.000	0.079***	0.000
Bachelor's degree	0.170***	0.000	0.173***	0.000
Other controls	Yes		Yes	
Adjusted R^2	0.578		0.497	
Number Observations	15,374		19,648	
Unemployment Rate				
Associate degree	−0.003	0.652	−0.044***	0.000
Bachelor's degree	−0.154***	0.000	−0.096***	0.000
Other controls	Yes		Yes	
Adjusted R^2	0.458		0.192	
Number of Observations	15,374		19,648	

Control variables include all those in Table 3 (including a spatial lag of the dependent variable using queen contiguity)

*, **, *** significance at the 10%, 5% and 1%, respectively

$6,000 between 2012 and 2022 (after adjusting for inflation) for bachelor's degrees, there was no measurable difference for those with associate degrees (NCES, 2024). The second and third possible mechanisms are the relationships between educational attainment and both unemployment rates and labor force participation. Additional spatial lag regressions in each period show that the percentage of the population with a bachelor's degree has stronger negative ties to unemployment and stronger positive links to labor force participation than does the percentage for associate degrees (Table 6).

4 Conclusion

The relationship between education and poverty is well documented. Past research has reported increased educational access and attainment as major factors linked to poverty reduction (Belfield and Bailey 2011; Jepsen et al. 2014. Despite federal and state government investment in educational institutions and policy initiatives aimed at increasing educational access and attainment (Ladd 2012; Jepsen et al. 2014; Strumbos et al. 2018), poverty continues to be prevalent in the USA. Some researchers have recently proposed increased investments in associate degree programs (Strumbos et al. 2018). Although the literature reports significant impacts of educational attainment on income, it lacks empirical evidence of how these relationships vary across space and time. The objective of this study was to determine the structural change and the spatial variation of the effects of education on poverty across seven states in the southern USA. Providing statistical evidence of the potential

effects of education across space and time will help policymakers formulate better educational initiatives aimed at poverty reduction.

Census tract-level data was compiled for 2010–2014 and 2017–2021 5-Year American Community Survey (ACS) for states typically considered as comprising the Deep South: Alabama, Florida, Georgia, Louisiana, Mississippi, Texas, and South Carolina. A multiscale geographically weighted regression (MGWR) that estimates a parameter for each census tract was used to determine spatial variations in the effects of education on poverty for each period. Maps of significant coefficients for the proportion of residents with associate and bachelor's degrees highlight tracts where these educational attainment levels had meaningful negative relationships with poverty.

Standardized spatial lag model results revealed strong and statistically significant relationships between poverty and high school degree, associate degree, and bachelor's degree for the 2010–2014 data in the Deep South. However, the magnitude and the significance of these relationships was generally smaller during the latter 2017–2021 period. This result supports our hypothesis on structural change in the effect of associate degree on poverty. Associate degree attainment demonstrated a negative relationship with poverty in the earlier period that has dissipated over time—even as the overall rate of attainment increased from 7.3 to 8.5 percent. Similarly, the relationship between poverty and bachelor's degree attainment was slightly lower over time despite the approximately 2.4% increase in the population with bachelor's degree over the decade. The control variables performed as expected, with unemployment rates and family structure (married / single-mother households) having the strongest relationships with poverty rates.

The results of the multiscale geographically weighted regression (MGWR) reveal more nuance to these global results. While the spatial lag results showed the coefficients of associate and bachelor's degrees reducing over time, the MGWR results demonstrate that the impacts of a bachelor's degree has become broader. Alternatively, the negative relationship between poverty rates and associate degree attainment is waning over time, with coefficients that are only significant in 18.8% of tracts for the 2017–2021 period relative to 54.4% in the 2010–2014 period. Poverty's inverse relationship with bachelor's degree attainment is moving in the opposite direction: while only statistically significant for 49% of tracts in 2010–2014, it increased to 63.3% in 2017–2021. Although not emphasized here, the relationship between a high school degree and poverty rates only held for 2.64% of the tracts in the 2017–2021 period compared to 17.5% in the earlier period, suggesting a dramatically weakened importance of a high school degree.

Is a college degree worth it despite the high cost of student debt? While this study does not exclusively analyze student debt in relation to the value of college degrees, the impact of college degrees evidenced in this study reveals significant value of four-year degrees on reducing poverty. Although the strength of the relationship has reduced over time, the impact still exceeds that of associate degree. College graduates continue to enjoy significantly higher lifetime earnings compared to their non-college counterparts. In addition, bachelor's degree holders have gained access to better-paying jobs and have greater job security. In fact, 95% of jobs created since the recession have required some college experience

(Sciaudone 2024). Although the rising cost of college education and the associated debt is a valid concern, the long-term financial and career benefits still make college education a worthy investment. However, personal goals, career aspirations, and the type of degree are important factors if the benefits of college education are to be realized. An important limitation of the approach in this paper is that it does not adequately address potential endogeneity issues—namely that areas with low poverty may increasingly attract college graduates. While some research on this topic has shown that industrial composition (controlled for here) and natural amenity endowment are some of the most important determinants of migration for both poor and non-poor (Nord 1998; Weinstein et al. 2023), future work may want to consider alternative strategies such as two-stage approaches that can address the endogeneity issue. No research that we are aware of has used instrumental variables in the multiscale geographically weighted regression framework.

While the econometric techniques used here do not lend themselves to firm claims of causality, the dramatic shifts demonstrated in Figs. 4 and 5 show clear changes in the relationship between education and poverty over a roughly 10-year period in the states in the Deep South. They also have implications for both educational policies and poverty reduction strategies. The results call for attention and a conscious effort by various stakeholders and policy makers at both the federal and state levels to enact educational policies that deliberately encourage, empower, and increase access to bachelor's degree education in the Deep South. Policymakers should prioritize initiatives that expand access to four-year degree programs, particularly in regions where the influence of bachelor's degree on reducing poverty has grown stronger over time. This could be in the form of county- or region-specific scholarships, alternative approaches to student loans for four-year degrees, or stronger linkages between two- and four-year degrees. In addition, the spatial nature of the education-poverty relationship revealed here suggests that educational attainment policies must be region-specific, focusing on local economic conditions and ensuring that educational programs are aligned with the unique needs of each county or state. Moreover, to maximize the poverty-reducing impact of education, policies should focus on creating stronger partnerships between educational institutions and local industries to ensure that degree programs prepare students for high-demand, well-paying jobs. Structural economic reforms to reduce unemployment through job creation, and educational and training opportunities for single-mother households (which saw improvements over the period analyzed here) may lead to greater income gains and subsequent poverty reduction. Such reforms should be directed toward addressing county-specific education and poverty challenges if results are to be realized.

Future research could attempt to replicate these results in other regions, or further explore the underlying reasons for the changing education–poverty relationships uncovered here. Quantifying the investments made in associate degree programs vs. bachelor's degree programs over this time may help shed light on why one approach became more effective at poverty prevention while the other floundered. Detailed data on community college facility characteristics, meshed with data on local industrial diversity, could also help explain why the associate

degree relationship changed so markedly over time. Regional science literature can make important contributions on this topic, especially with cutting-edge tools like MGWR that allow for the estimation of local relationships.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

Albrecht DE, Albrecht SL (2000) Poverty in nonmetropolitan America: impacts of industrial, employment, and family structure variables. Rural Sociol 65(1):87–103

Alesina A, Baqir R, Easterly W (1999) Public goods and ethnic divisions. Q J Econ 114(4):1243–1284

Anselin L (2001) Spatial econometrics. A compainion to Theoretical Econometrics, 310330

Atkinson A (1992) Measuring poverty and differences in family composition. Economica 59(233):1–16

Awan MS, Malik N, Sarwar H, Waqas M (2011) Impact of education on poverty reduction. Int J Acad Res 3(1):659–664

Bailey TR, Kienzl GS, Marcotte D (2004) The return to a sub-baccalaureate education: The effects of schooling, credentials, and program of study on economic outcomes. Report prepared for National Assessment of Vocational Education, U.S. Department of Education

Baker R (2022) The historical racial regimen and racial inequality in poverty in the American South. Am J Sociol 127(6):1721–1781

Baum S, Ma J, Payea K (2010) Education Pays, 2010: The Benefits of Higher Education for Individuals and Society. Trends in Higher Education Series. College Board Advocacy & Policy Center

Belfield CR, Bailey T (2011) The benefits of attending community college: a review of the evidence. Commun Coll Rev 39(1):46–68

Blank R (2005) Poverty, policy, and place: how poverty and policies to alleviate poverty are shaped by local characteristics. Int Reg Sci Rev 28(4):441–464

Brucker DL, Houtenville AJ (2015) People with disabilities in the United States. Arch Phys Med Rehabil 96(5):771–774

Brunsdon C, Fotheringham S, Charlton M (2002) Geographically weighted regression. J Royal Stat Soc: Series D (Stat) 47(3):431–443

Buchmann C, Brakewood D (2000) Labor structures and school enrollments in developing societies: Thailand and Kenya compared. Comp Educ Rev 44:175–204

Bureau of Labor Statistics. 2024a. Median Usual Weekly Earnings of Full-time Wage and Salary Workers by Educational Attainment. Available online: https://www.bls.gov/charts/usual-weekly-earnings/usual-weekly-earnings-over-time-by-education.htm

Bureau of Labor Statistics. 2024b. Occupational Outlook Handbook. Available online. https://www.bls.gov/ooh/construction-and-extraction/plumbers-pipefitters-and-steamfitters.htm#tab-1

Cancian M, Haskins R (2014) Changes in family composition: Implications for income, poverty, and public policy. Ann Am Acad Pol Soc Sci 654(1):31–47

Carnevale, A. and N. Smith. (2022). Training programs are welcome, but let's not overlook the benefits of a bachelor's degree. Hechinger Report. Available online: https://hechingerreport.org/opinion-training-programs-are-welcome-but-lets-not-overlook-the-benefits-of-a-bachelors-degree/

Chan R (2016) Understanding the purpose of higher education: An analysis of the economic and social benefits for completing a college degree. Journal of Education Policy, Planning and Administration 6(5):1–40

Colclough C, Rose P, Tembon M (2000) Gender inequalities in primary schooling: the roles of poverty and adverse cultural practice. Int J Educ Dev 20:5–27
Cromartie J, Swanson L (1996) Census tracts more precisely define rural populations and areas. Rural Dev Perspect 11(3):31–39
Cruse L, Milli J, Contreras-Mendez S, Holtzman T, Gault B (2022) Investing in Single Mothers' Higher Education in South Carolina. Institute for Women's Policy Research, Publication R641
Dadgar M, Trimble MJ (2015) Labor market returns to sub-baccalaureate credentials: How much does a community college degree or certificate pay? Educ Eval Policy Anal 37(4):399–418
Deshpande N (2021) An evolving role for college. *Econ Focus*, Fourth Quarter, 8–11. https://bit.ly/college-alternative8
Fallows J (2014) Theories of History: Joe Max Higgins and the Golden Triangle of Mississippi. The Atlantic. Available online: https://www.theatlantic.com/business/archive/2014/05/joe-max-higgins-and-the-making-of-the-golden-triangle/371550/
Farber HS (2004) Job loss in the United States, 1981–2001. In Accounting for Worker Well-Being. Emerald Group Publishing Limited, Bingley, pp 69–117
Fotheringham AS, Yang W, Kang W (2017) Multiscale geographically weighted regression (MGWR). Ann Am Assoc Geogr 107(6):1247–1265
Fry R, Braga D, Parker K (2024) Is College worth it? https://www.pewresearch.org/social-trends/2024/05/23/is-college-worth-it-2/
Gallardo R, Whitacre B (2018) 21st century economic development: telework and its impact on local income. Reg Sci Policy Pract 10:103–123
Hanratty M, Blank R (1993) Down and out in North America: recent trends in poverty rates in the US and Canada. Q J Econ 10:233–257
Haskins R, Holzer HJ, Lerman R (2009) Promoting economic mobility by increasing postsecondary education. Report for Pew Charitable Trusts' Economic Mobility Project. Available online: https://www.pewtrusts.org/en/research-and-analysis/reports/0001/01/01/promoting-economic-mobility-by-increasingpostsecondary-education
Hout M (2012) Social and economic returns to college education in the United States. Ann Rev Sociol 38:379–400
Jacobson L, LaLonde R, Sullivan DG (2005) Estimating the returns to community college schooling for displaced workers. J Econ 125(1–2):271–304
Jepsen C, Troske K, Coomes P (2014) The labor-market returns to community college degrees, diplomas, and certificates. J Law Econ 32(1):95–121
Karahasan B (2023) Do new firms boost local innovation? Evidence from Turkey. Int Reg Sci Rev. https://doi.org/10.1177/01600176231173433
Kures M, Deller S (2023) Growth in commuting patterns and their impacts on rural workforce and economic development. Econ Dev Q 37(1):54–63
Ladd HF (2012) Education and poverty: confronting the evidence. J Policy Anal Manage 31(2):203–227
Lahr M (2020) Poverty, the U.S. South, and the SRSA. Rev Reg Stud 50(2):153–169
Lavrinovicha I, Lavrinenko O, Teivans-Treinovskis J (2015) Influence of education on unemployment rate and incomes of residents. Procedia Soc Behav Sci 174:3824–3831
Levernier W, Partridge MD, Rickman DS (2000) The causes of regional variations in US poverty: a cross-county analysis. J Reg Sci 40(3):473–497
Li Z, Fotheringham S (2020) Computational Improvements to multi-scale geographically weighted regression. Int J Geogr Inf Sci 34(7):1378–1397
Liu W, Li J, Zhao R (2023) The effects of rural education on poverty in China: a spatial econometric perspective. J Asia Pac Econ 28(1):176–198
Ma J, Pender M, Welch M (2016) Education Pays 2016: The Benefits of Higher Education for Individuals and Society. Trends in Higher Education Series. College Board
MacDonald G, Pettit K (2019) Open, accessible data on jobs and workers: Tract-Level LODES Data. Available online: https://urban-institute.medium.com/open-accessible-data-on-jobs-and-workers-tract-level-lodes-data-945fcac9e280
Morgan J, David M (1963) Education and income. Q J Econ 77:423–437
National Center for Education Statistics (2024) Annual Earnings by Educational Attainment, 2012 through 2022. Available online: https://nces.ed.gov/programs/coe/indicator/cba/annual-earnings
Nord M (1998) Poor people on the move: County-to-County migration and the spatial concentration of poverty. J Reg Sci 38(2):329–351

Padda IUH, Hameed A (2018) Estimating multidimensional poverty levels in rural Pakistan: a contribution to sustainable development policies. J Clean Prod 197:435–442
Palmer J (2011) Disability and poverty: a conceptual review. J Disabil Policy Stud 21(4):210–218
Partridge MD, Rickman DS (2005) High-poverty nonmetropolitan counties in America: can economic development help? Int Reg Sci Rev 28(4):415–440. https://doi.org/10.1177/0160017605278998
Partridge MD, Rickman DS (2006) The geography of American poverty: is there a need for place-based policies? Kalamazoo, WE Upjohn Institute
Partridge MD, Rickman DS (2007) Persistent pockets of extreme American poverty and job growth: Is there a place-based policy role? J Agric Resour Econ 32:201–224
Rank MR, Eppard LM, Bullock HE (2021) Poorly Understood: What America Gets Wrong about Poverty. Oxford University Press, Oxford
Riddell WC, Song X (2011) The impact of education on unemployment incidence and re-employment success: evidence from the US labour market. Labour Econ 18(4):453–463
Rupasingha A, Goetz SJ (2007) Social and political forces as determinants of poverty: a spatial analysis. J Socio-Econ 36(4):650–671
Saunders P (2002) The direct and indirect effects of unemployment on poverty and inequality. Aust J Labour Econ 5(4):507–529
Sachdeva M, Fotheringham S, Li Z (2022) Do places have value? Quantifying the intrinsic value of housing neighborhoods using MGWR. J Hous Res 31(1):24–52
Sciaudone C (2024) Is college worth it? A balanced look at the Pros and Cons. Business Insider. https://www.businessinsider.com/personal-finance/student-loans/is-college-worth-it
Šileika A, Bekerytė J (2013) Theoretical issues of the relationship between unemployment, poverty, and crime in sustainable development. J Secur Sustain Issues 2:59–70
Slack T, Jensen L (2004) Employment adequacy in extractive industries: an analysis of underemployment, 1974–1998. Soc Nat Resour 17(2):129–146
Slack T, Singelmann J, Fontenot K, Poston DL Jr, Saenz R, Siordia C (2009) Poverty in the Texas borderland and lower Mississippi Delta: a comparative analysis of differences by family type. Demogr Res 20:353–376
Strumbos D, Linderman D, Hicks CC (2018) Postsecondary pathways out of poverty: City University of New York accelerated study in associate programs and the case for national policy. RSF: the Russell Sage Foundation Journal of the Social Sciences 4(3):100–117
Snyder AR, McLaughlin DK (2004) Female-headed families and poverty in rural America. Rural Sociol 69(1):127–149
Snyder AR, McLaughlin DK, Findeis J (2006) Household composition and poverty among female-headed households with children: differences by race and residence. Rural Sociol 71(4):597–624
Tierney WG (2015) Rethinking education and poverty. JHU Press, Baltimore
Tough P (2023). Americans are losing Faith in the value of College. Whose Fault is that? The New York Times. https://www.nytimes.com/2023/09/05/magazine/college-worth-price.html?smid=url-share
Tsaliki PV (2009) Economic development and unemployment: do they connect? Int J Soc Econ 36(7):773–781
Ukpere WI, Slabbert AD (2009) A relationship between current globalization, unemployment, inequality, and poverty. Int J Soc Econ 36(1/2):37–46
Watson P, Deller S (2022) Tourism and economic resilience. Tour Econ 28(5):1193–1215
Weber B, Jensen L, Miller K, Mosley J, Fisher M (2005) A critical review of the rural poverty literature: is there truly a rural effect? Int Reg Sci Rev 28(4):381–414
Weinstein AL, Hicks M, Wornell E (2023) An aggregate approach to estimating quality of life in micropolitan areas. Ann Reg Sci 70(2):447–476
Yu H, Fotheringham A, Li Z, Oshan T, Kang W, Wolf L (2020) Inference in multiscale geographically weighted regression. Geogr Anal 52(1):87–106
Zhang L, Liu X, Hu Y (2024) Degrees of return: estimating internal rates of return for college majors using quantile regression. Am Educ Res J 61(3):577–609

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.